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प्रयागराज–211004 भारत
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Computer Science and Engineering Department
End Semester (Odd) Examination 2023-24

Programme Name: B.Tech	**Semester: V**
Course Code: CSN15403	**Course Name: Cryptography and Network Security**
	Student Reg. No.:
Duration: 150 Minutes	**Max. Marks: 50**

Instructions: (Related to Questions)

1. *Precise answer with proper justification will be considered to get complete marks.*
2. *Attempt all questions.*

			Marks
Q1	**a.**	Although RSA and ElGamal are secure asymmetric-key cryptosystems, their security comes with a price, their large keys. For instance, ECC over the GF(2^n) with n of 160 bits can provide the same level of security as RSA with n of 1024 bits Justify the above statement by demonstrating the elliptic curve cryptosystem process of encryption and decryption.	**10**
	b.	Further, with elliptic curve E(1, 2) over the GF(11) field, solve the following questions i. For the point e_1 = (2, 1) and d = 3, generate public and private keys for Bob. ii. For the point on the curve P = (4, 2) as a plaintext for Alice, create ciphertext corresponding to the plaintext with r as 6.	
Q 2	**a.**	The Digital Signature Standard (DSS) was adopted by the National Institute of Standards and Technology (NIST) in 1994. NIST published DSS as FIPS 186. However, the signing and verification process of DSS criticized from the time it was	**10**

	b.	published due its design. Justify your view by explaining the DSS key generation, signing process and verification process. Using the DSS scheme, let q = 59, p = 709, and d = 14. Find values for e1 and e2. Choose r = 13. Find the value of S1 and S2 if h(M) = 100. Verify the signature.	
Q 3	**a.** **b.**	In password authentication, the claimant needs to send her secret (the password) to the verifier; this is subject to eavesdropping by the attacker. In addition, a dishonest verifier could reveal the password to others or use it to impersonate the claimant. Propose a solution for this scenario and justify Fiat-Shamir method with necessary proof. For p = 569, q = 683, and s = 157, show two rounds of the Fiat-Shamir protocol with challenges c=0 and c=1 by calculating the values (as discussed in the class n, v, x, y) with respect to r=203,122 and r=153,271.	**10**
Q4	**a**		
Q5	**a**		
Q6	**a**		